Linear Regression and Beyond

Lecture 02

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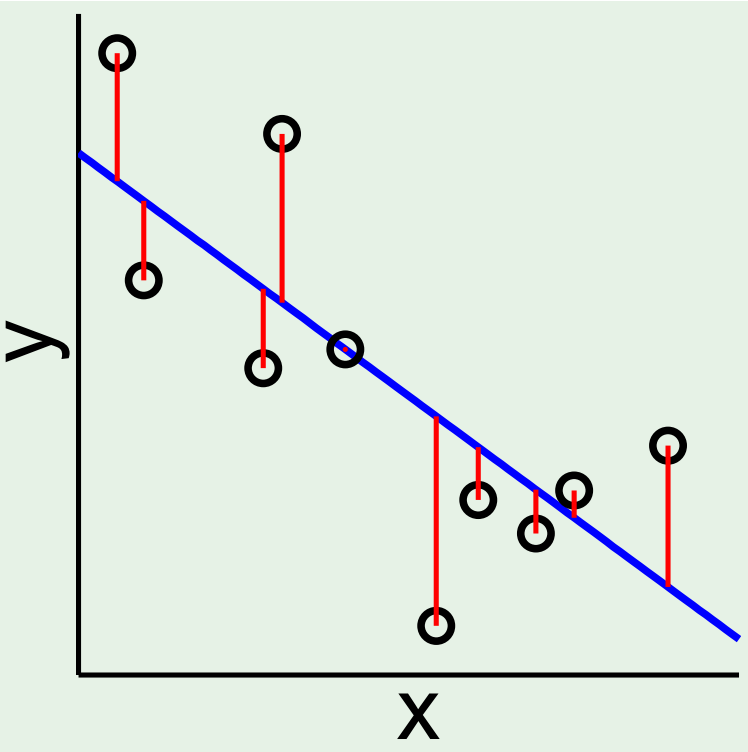
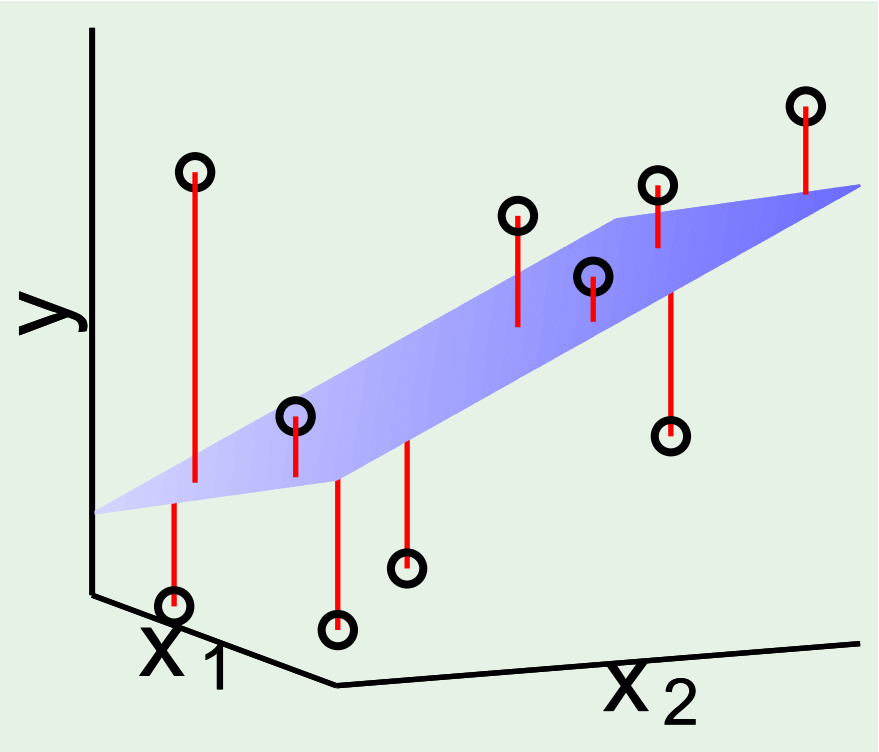
Outline

- **Linear regression**
  - Definition and derivation
- Extensions and beyond
- Regularization
  - On linear regression
  - On others

Linear regression

"Regression" usually means "real-value output".



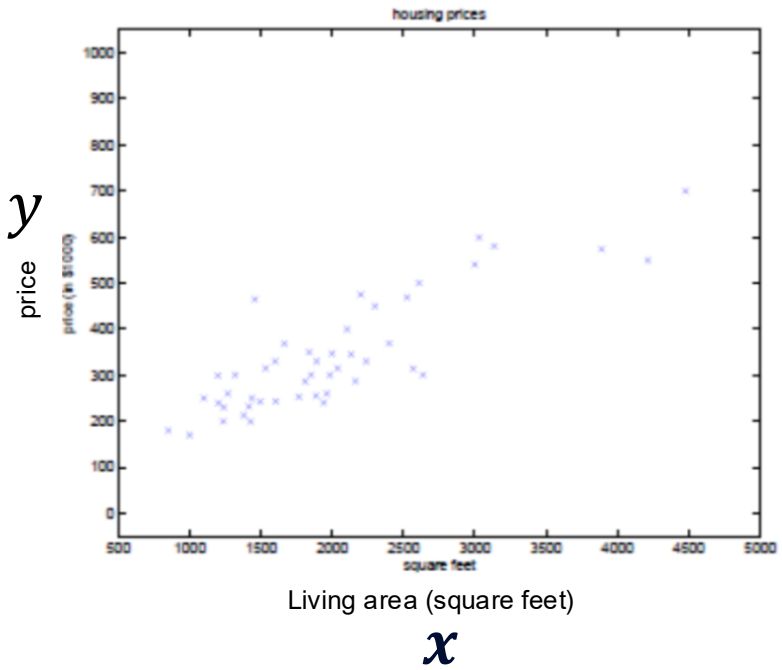


A learning problem of house hunting

- Suppose we have a dataset giving the living areas and prices of some houses :

	Living area (feet^2)	Price(1000$s)	
$\boldsymbol{x}_1$	2014	400	y_1
$\boldsymbol{x}_2$	1600	330	y_2
$\vdots$	2400	369	$\vdots$
	1416	232	
$\boldsymbol{x}_t$	3000	540	y_t
$\vdots$	...		$\vdots$
	2005	?	
	3200	?	
	1280	?	

We can plot this data set:



- How can we learn to predict the prices (y) of other houses, as a function of the size of their living areas ($\boldsymbol{x}$)?

(From Andrew Ng's slides)

A slightly richer dataset

- If you want to find the most reasonably priced house satisfying your needs: square-ft, # of bedroom, distance to workplace...

$\boldsymbol{x}_t \in R^2$

Living area (feet²)	# bedrooms	Price(1000$s)
2014	3	400
1600	3	330
2400	3	369
1416	2	232
3000	4	540
…		
2005	3	?
3200	4	?
1280	2	?

$$\boldsymbol{x}_t = [x_{t1}, x_{t2}]^T$$

The learning problem

- **Features:**
  - Living area, #bedroom, distance to workplace …
  - Denote as $\boldsymbol{x} = [x_1, x_2, \dots, x_n]^T$, where n is number of features.

- **Target:**
  - Price
  - Denoted as y

$$X = [\boldsymbol{x}_1 \quad \cdots \quad \boldsymbol{x}_N] = \begin{bmatrix} x_{11} & \cdots & x_{1N} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nN} \end{bmatrix}$$

$$Y = [y_1 \quad \cdots \quad y_N]$$

- **Goal:**
  - Given a data set $D = \{(x_t, y_t), t = 1, \dots, N\}$, find a function $f(x|\theta)$ to predict y accurately.

Linear regression

Assume that y (target) is a linear function of $\boldsymbol{x}$ (features), i.e.,

$$f(\boldsymbol{x}|\theta) = a_0 + a_1 x_1 + a_2 x_2 + \cdots + a_n x_n$$

Without loss of generality, we may rewrite it as:

$$f(\boldsymbol{x}|\theta) = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = \boldsymbol{a}^T \boldsymbol{x}$$ (Let $x_1 = 0$)

$f(\boldsymbol{x}|\theta)$ is linear in $\boldsymbol{a}$.

Here, $\theta = \{a_1, \ldots, a_n\}$ are the parameters (also called weights) parameterizing the space of linear functions mapping from the feature space to the target space.

The Least Mean Square (LMS) method 最小二乘法

- Prediction error for the data $(\boldsymbol{x}_t, y_t)$ by the squared error (Euclidean distance)

$$||y_t - f(x_t|\theta)||^2$$

- Sum of the prediction error

$$J(D|\theta) = \sum_{t=1}^{N} ||y_t - f(x_t|\theta)||^2 = \sum_{t=1}^{N} ||y_t - \boldsymbol{a}^T \boldsymbol{x}_t||^2$$

- In the matrix form:

$$J(D|\theta) = (Y - a^T X)(Y - a^T X)^T$$

- Minimizing the prediction error leads to:

$$\boldsymbol{a} = (XX^T)^{-1} XY^T$$

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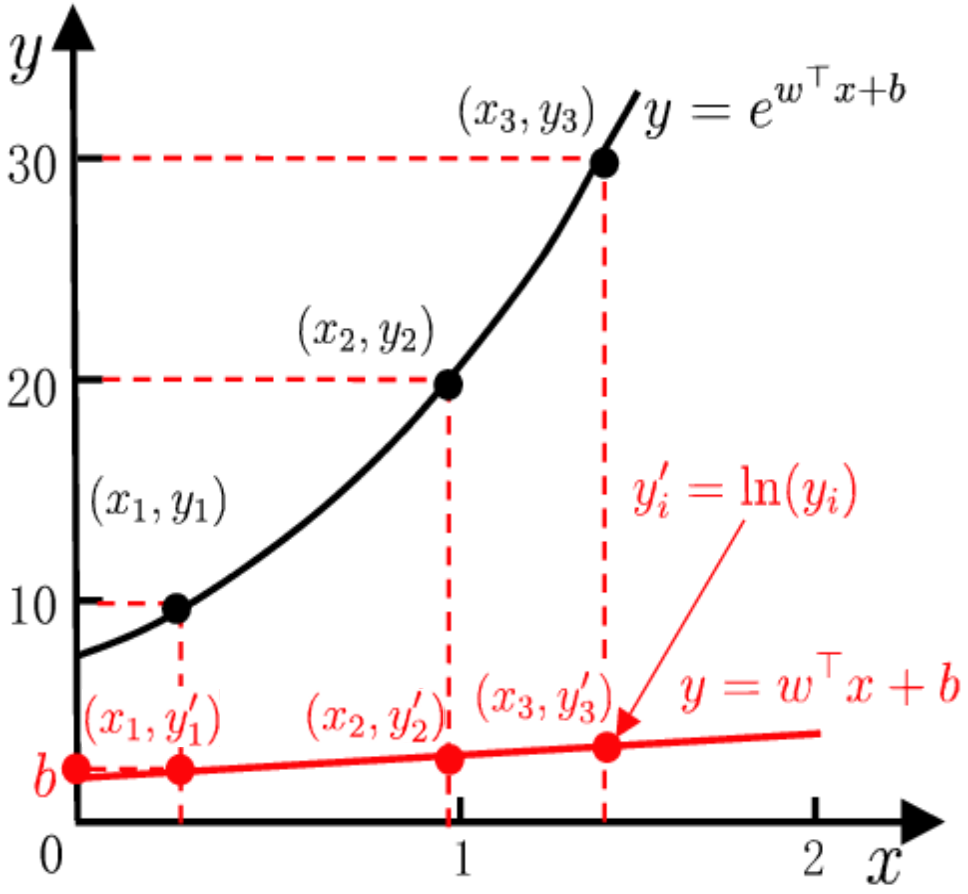
Log-linear model regression

- The log output is the target of the linear model



$$\ln y = \boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b$$

$$\Updownarrow$$

$$y = \boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b$$

Generalized linear model

- The general form

$$y = g^{-1}\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b\right)$$

- g( ) is called the link function, usually a monotone differentiable function

- Log-linear regression is a special case of generalized linear model $g(\cdot) = \ln(\cdot)$

Extension by Linear Basis Function

$$y(\boldsymbol{x}, \boldsymbol{w}) = \sum_{j=0}^{M-1} w_j \phi_j(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{\phi}(\boldsymbol{x})$$

$$\mathbf{w} = (w_0, \ldots, w_{M-1})^{\mathrm{T}}$$

$$\boldsymbol{\phi} = (\phi_0, \ldots, \phi_{M-1})^{\mathrm{T}}$$

$$\phi_0(\boldsymbol{x}) = 1$$

w_0 a bias parameter

Basis function choices

Polynomial $\phi_j(x) = x^j$

Gaussian $\phi_j(x) = \exp\left(-\frac{(x-\mu_j)^2}{2s^2}\right)$

Sigmoidal $\phi_j(x) = \sigma\left(\frac{x-\mu_j}{s}\right)$ with $\sigma(a) = \frac{1}{1+e^{-a}}$

splines, Fourier, wavelets, *etc.*

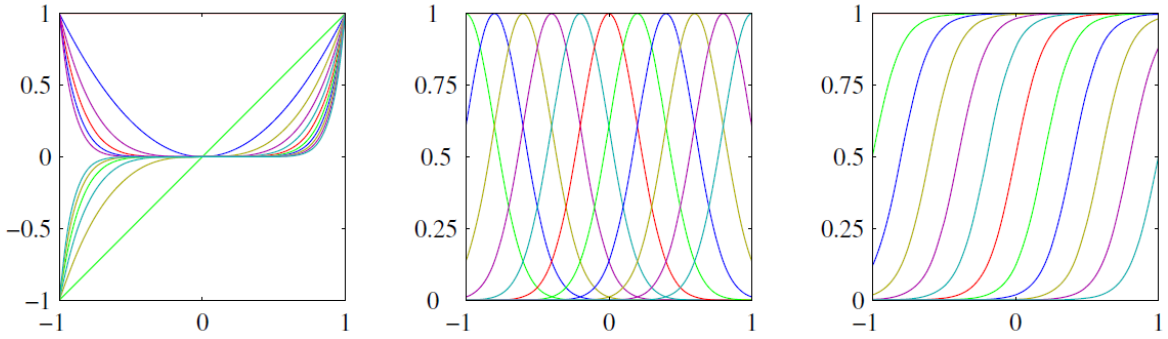
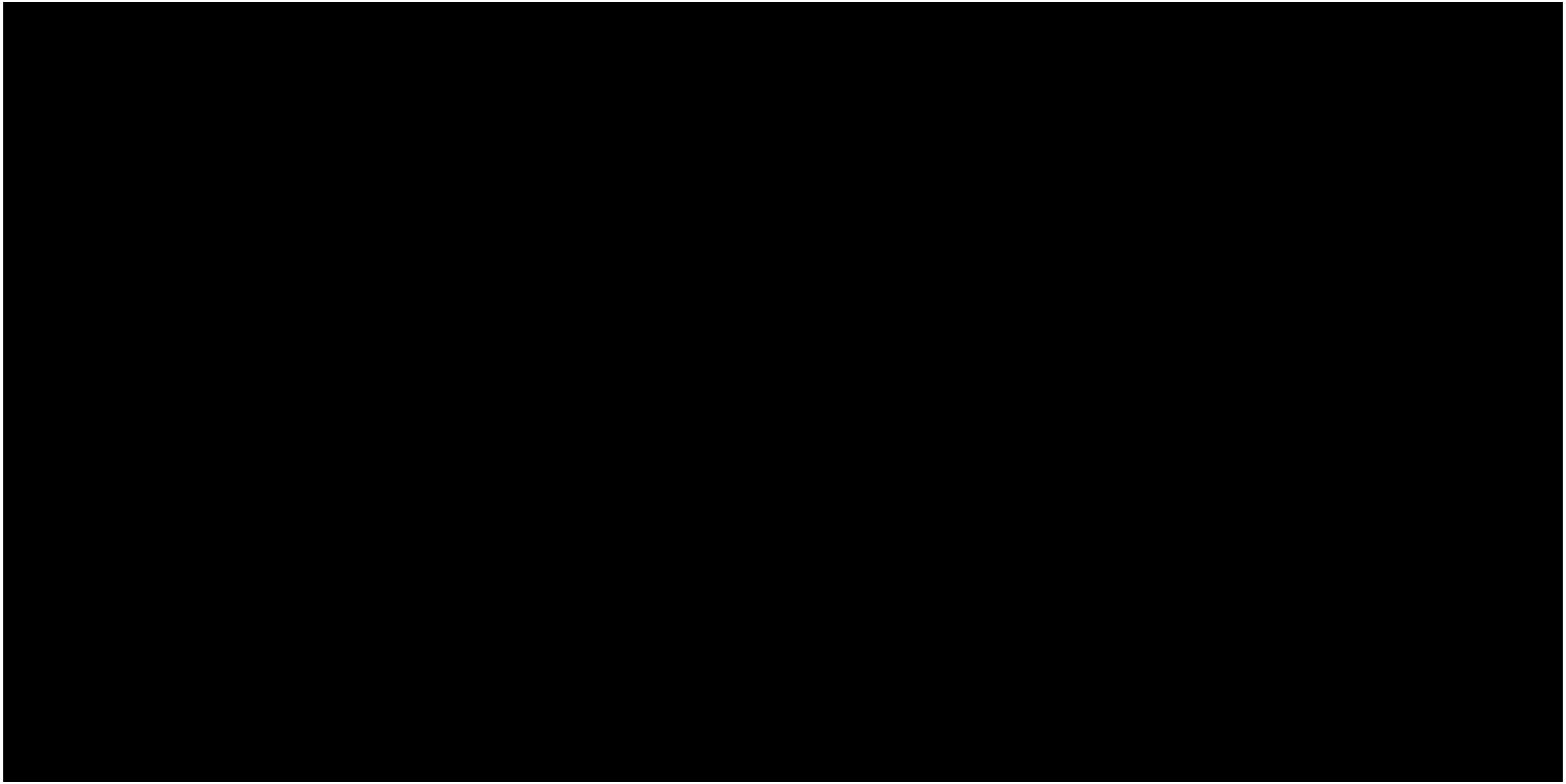


Figure 3.1 Examples of basis functions, showing polynomials on the left, Gaussians of the form (3.4) in the centre, and sigmoidal of the form (3.5) on the right.

Basis functions work as feature transformation



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Regularized least square

- Regularization has the advantage of limiting the model complexity. It is helpful for the cases when the training sample size is small, to prevent from overfitting.
- This is replaced with the problem of finding a suitable value of the regularization coefficient.

$$J(D|\theta) = \sum_{t=1}^{N} ||y_t - \boldsymbol{a}^T \boldsymbol{x}_t||^2 + \lambda ||\boldsymbol{a}||^2$$

- The corresponding optimal solution is

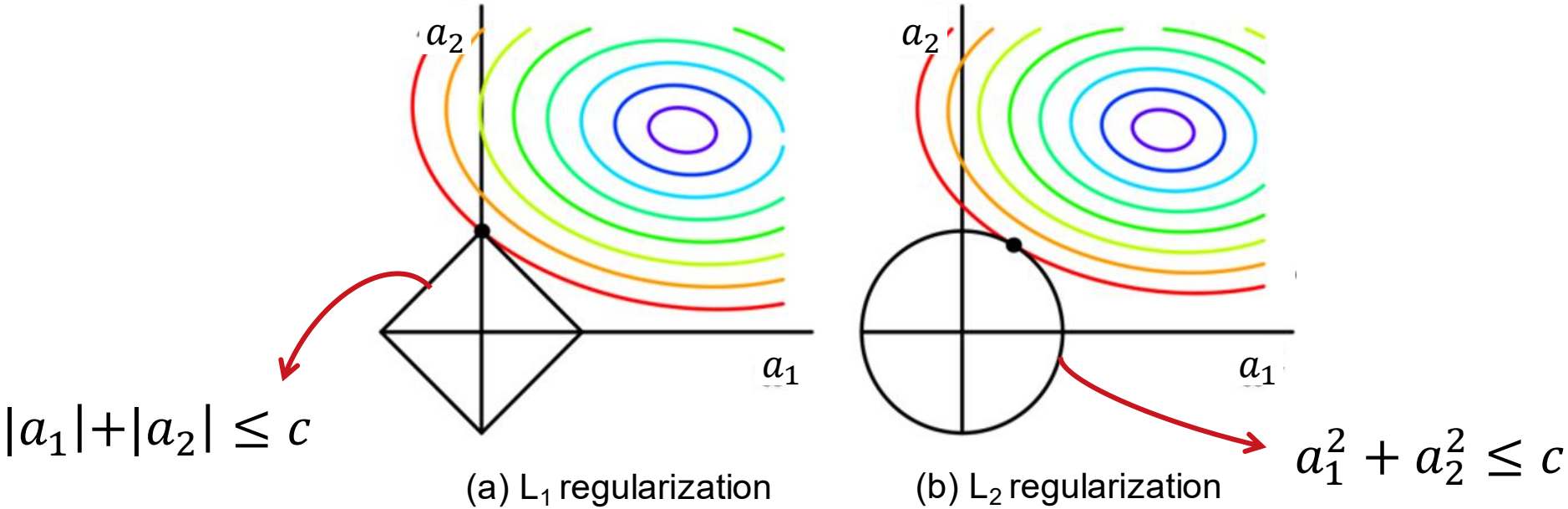
$$\boldsymbol{a} = (\lambda I + XX^T)^{-1} XY^T$$

- More general regularization form:

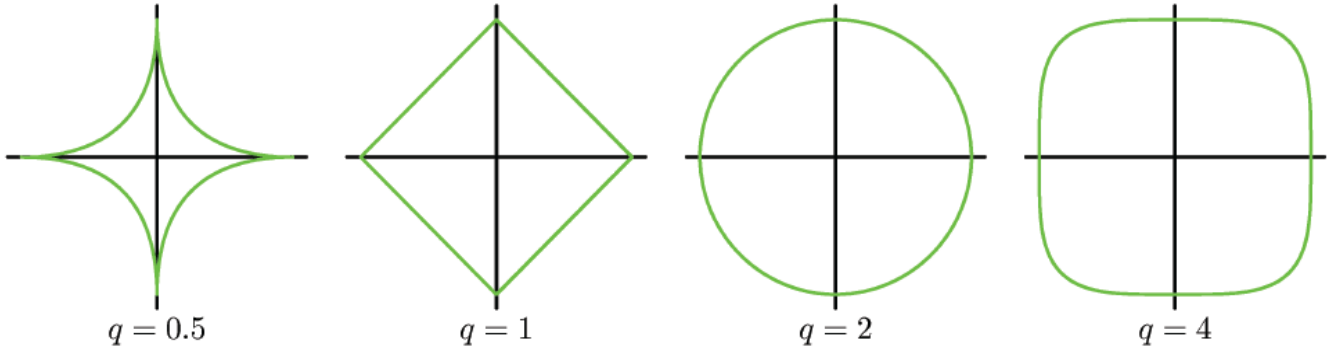
$$\lambda \sum_{i=1}^{n} |a_i|^q$$

Effects of different regulations

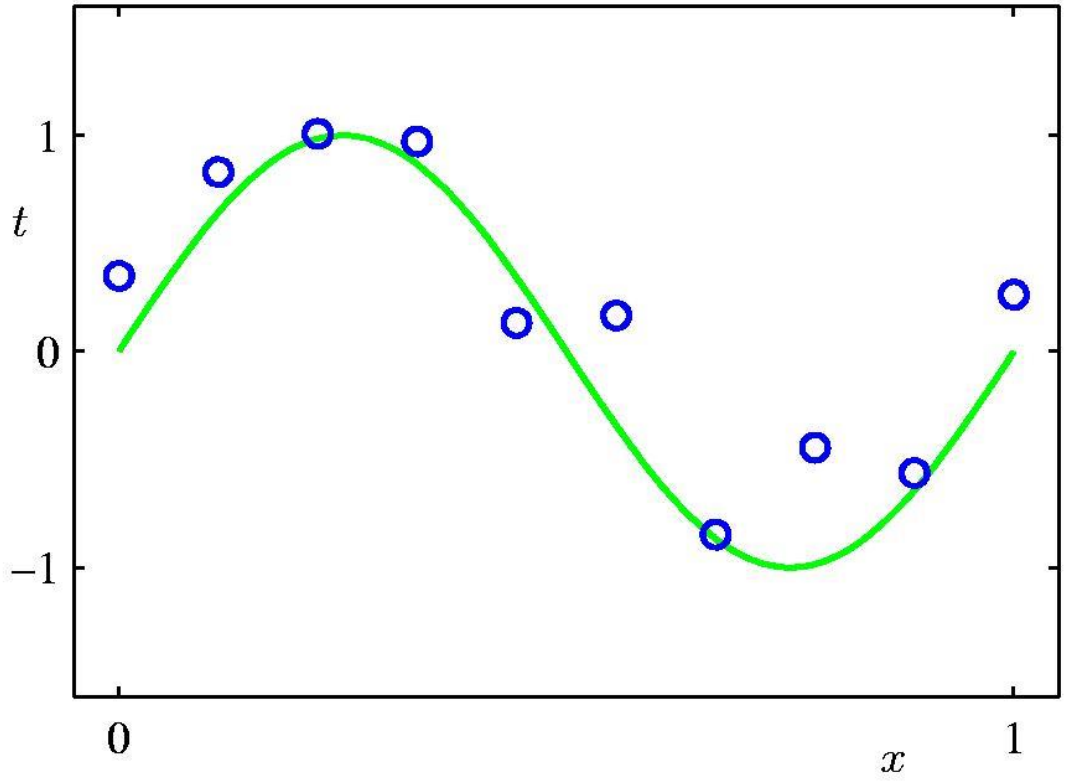
- L1-norm regularization tends to generate sparser solutions than L2-norm (i.e., quadratic) regularization.



(a) L_1 regularization (b) L_2 regularization

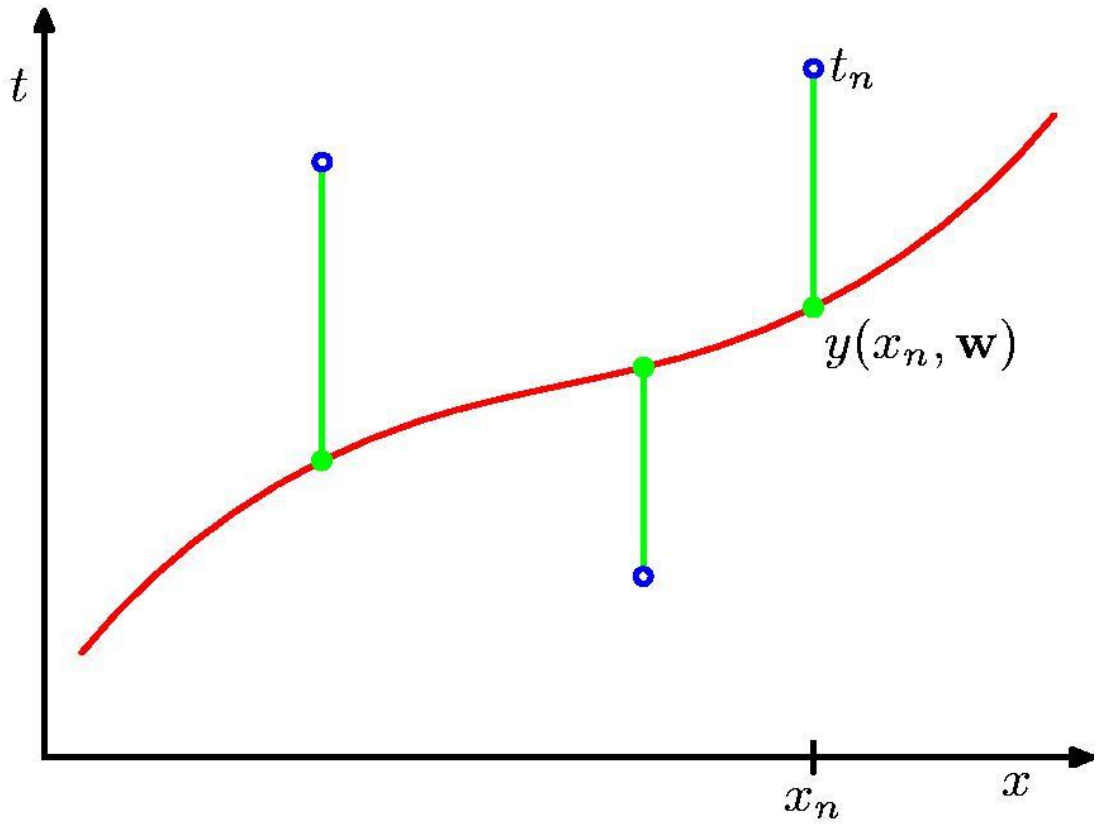


Polynomial Curve Fitting



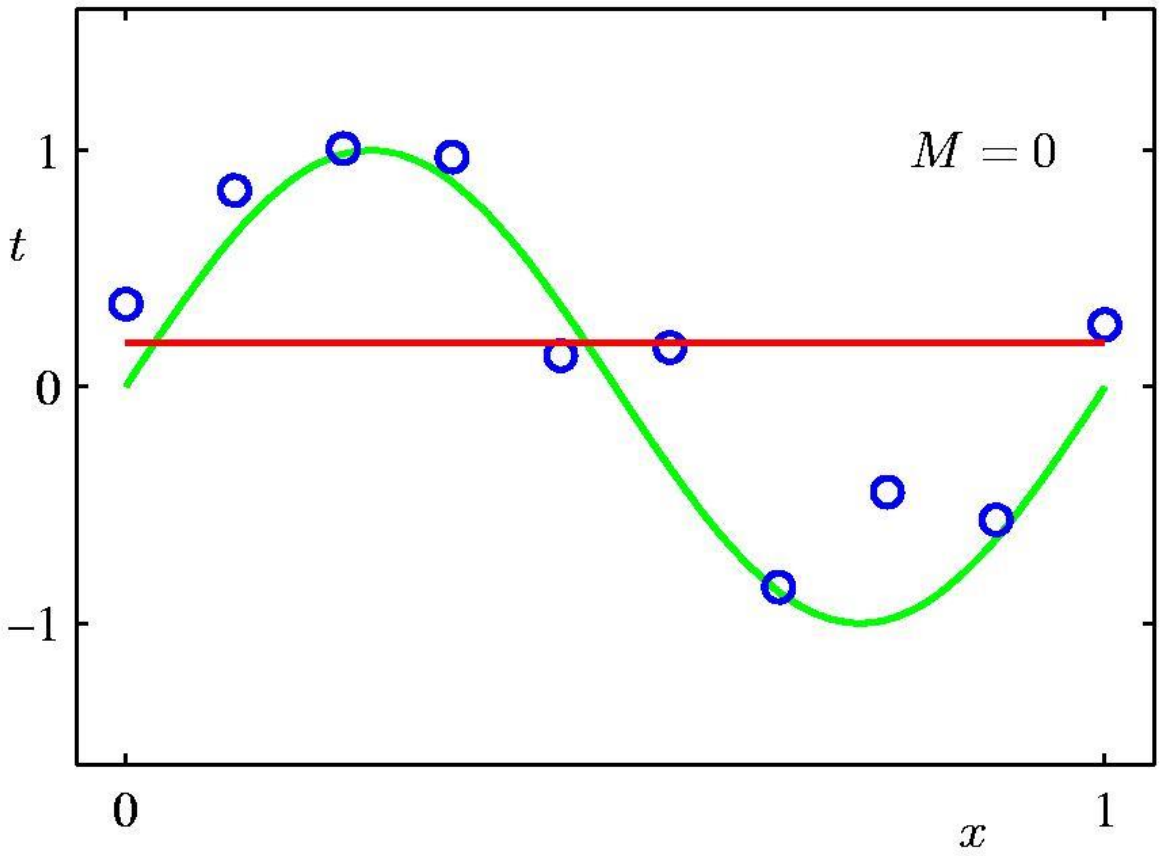
$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

Sum of square error



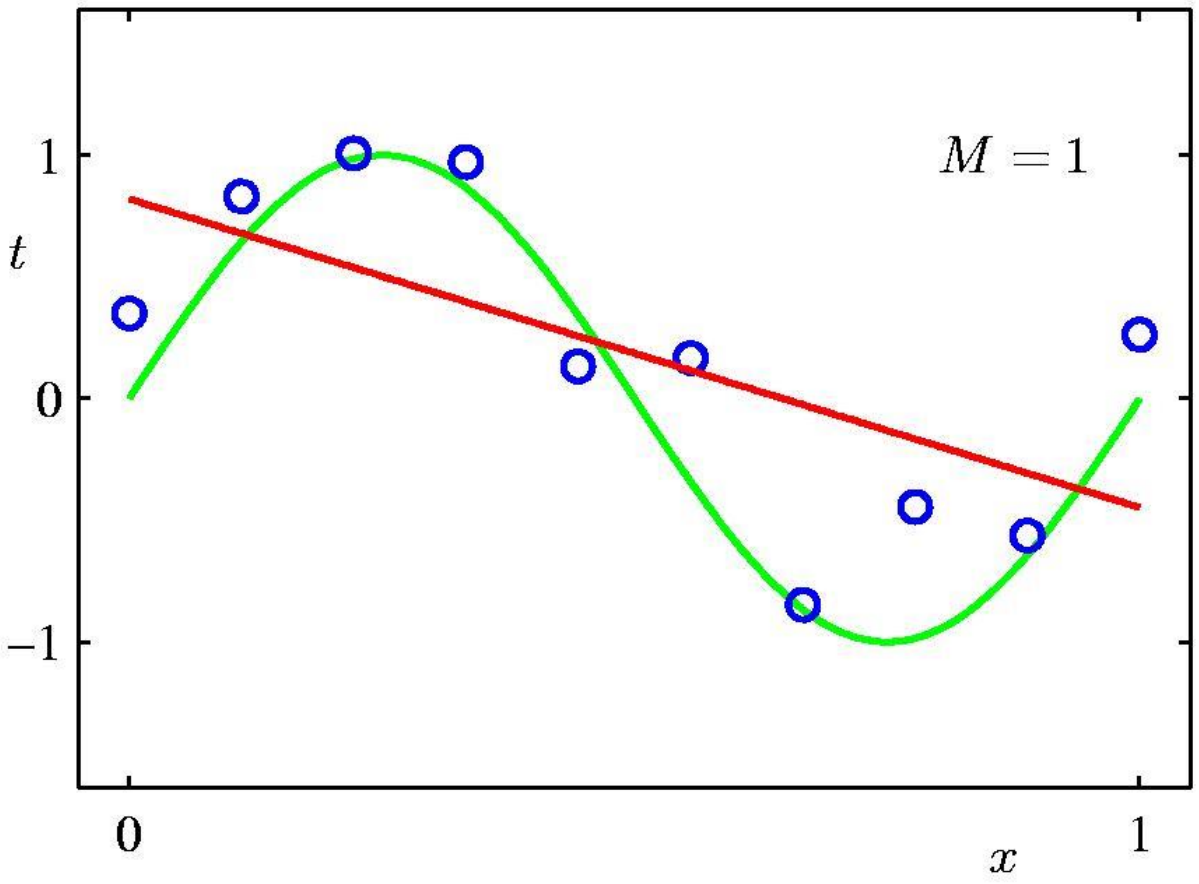
$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \mathbf{w}) - t_n\}^2$$

The 0-th Order Polynomial



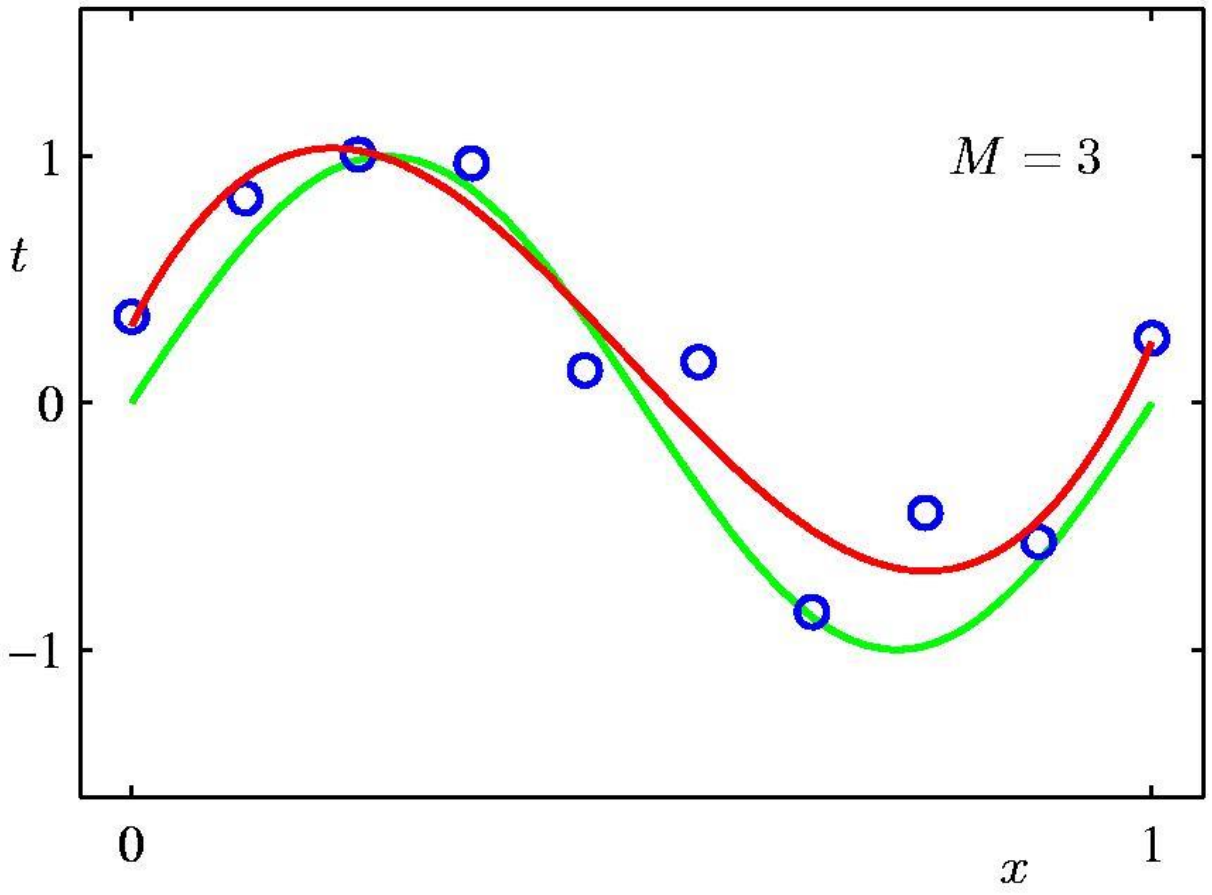
$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

The 1-st Order Polynomial



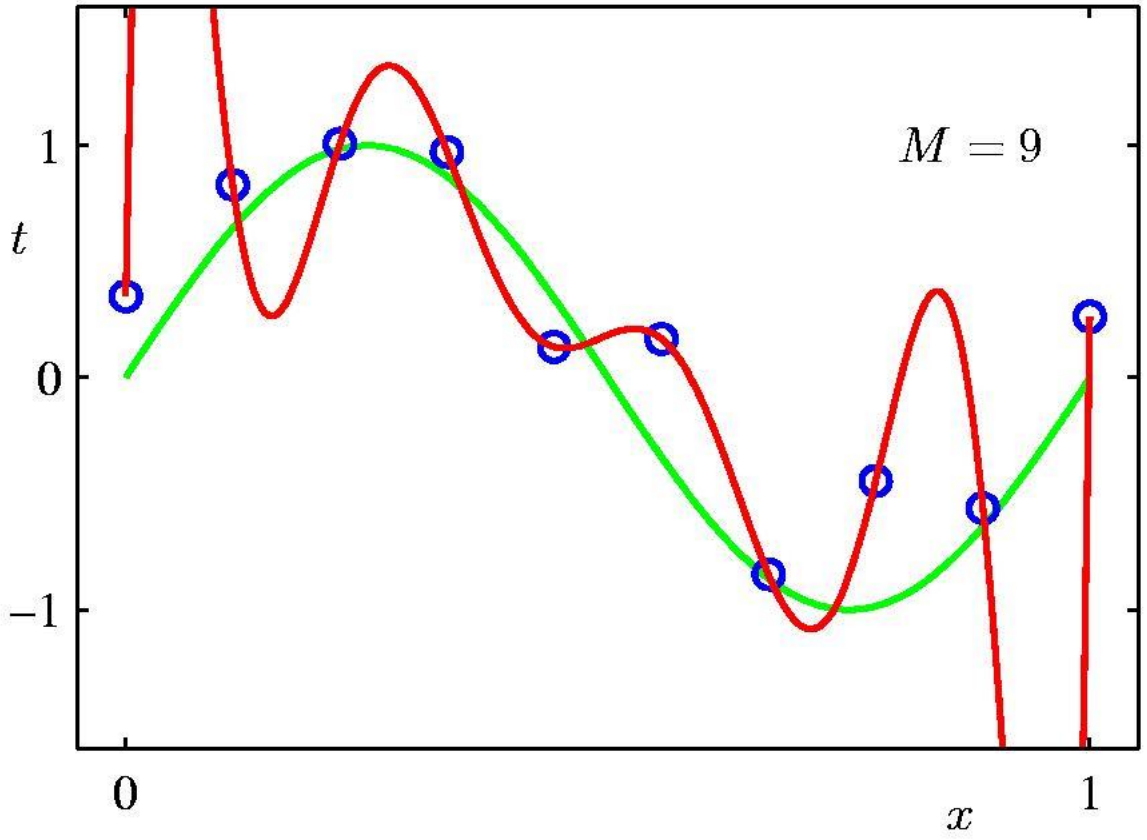
$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

The 3-rd Order Polynomial



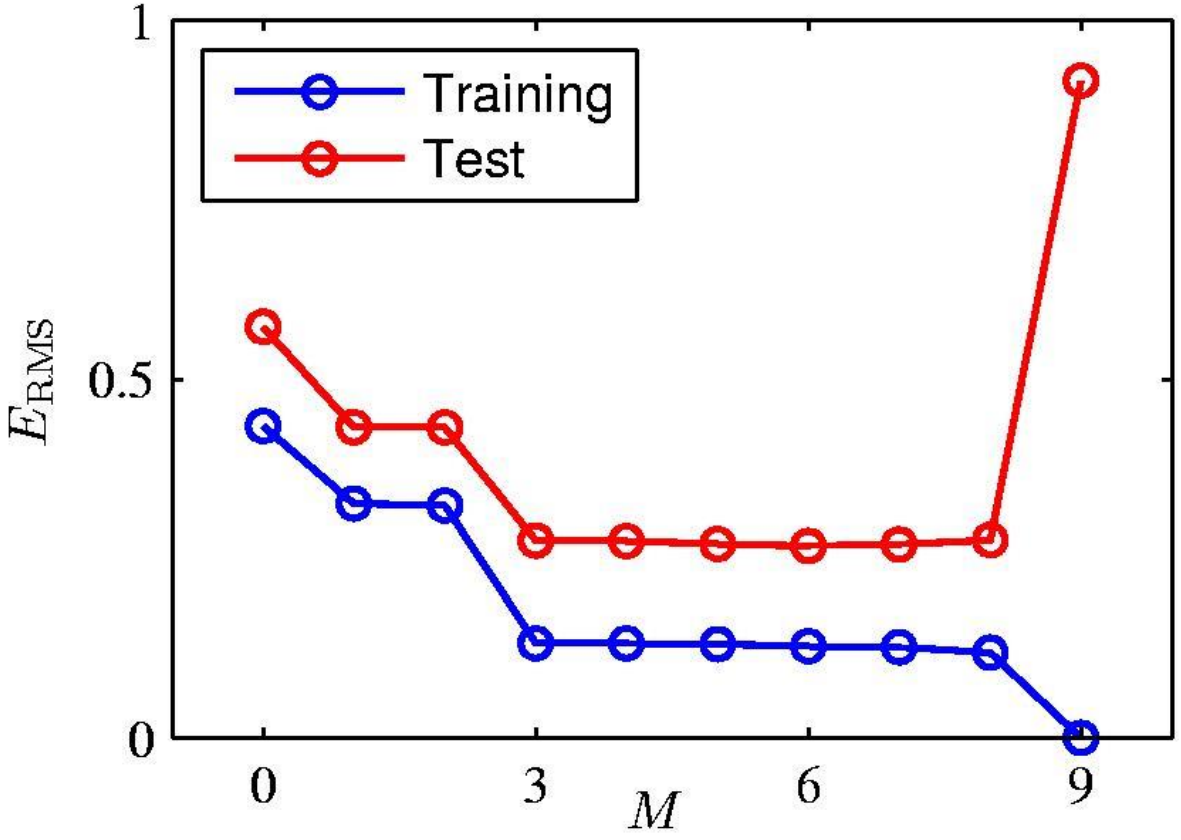
$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

The 9-th Order Polynomial



$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

Overfitting



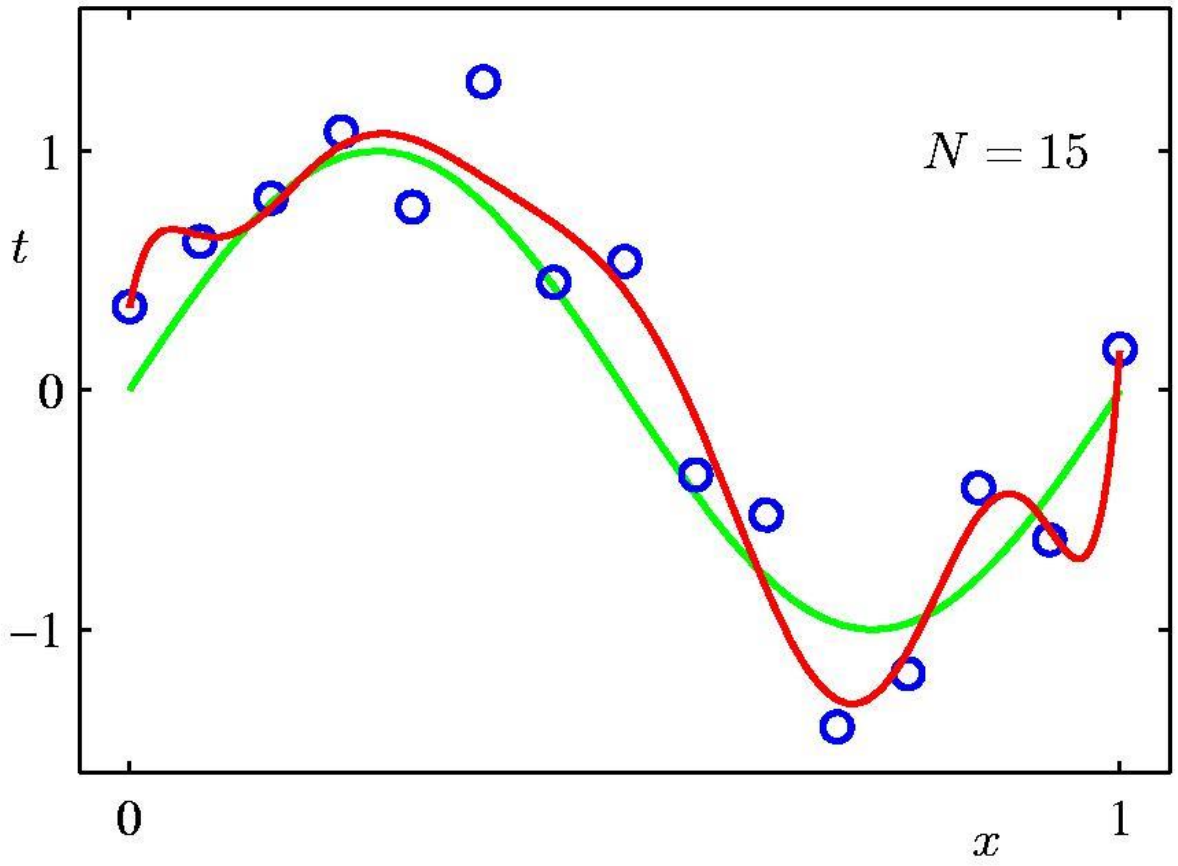
Root-Mean-Square (RMS) Error: $E_{\mathrm{RMS}} = \sqrt{2E(\mathbf{w}^\star)/N}$

Polynomial Coefficients

	$M=0$	$M=1$	$M=3$	$M=9$
$w_0^\star$	0.19	0.82	0.31	0.35
$w_1^\star$		-1.27	7.99	232.37
$w_2^\star$			-25.43	-5321.83
$w_3^\star$			17.37	48568.31
$w_4^\star$				-231639.30
$w_5^\star$				640042.26
$w_6^\star$				-1061800.52
$w_7^\star$				1042400.18
$w_8^\star$				-557682.99
$w_9^\star$				125201.43

When the training data set size N varies

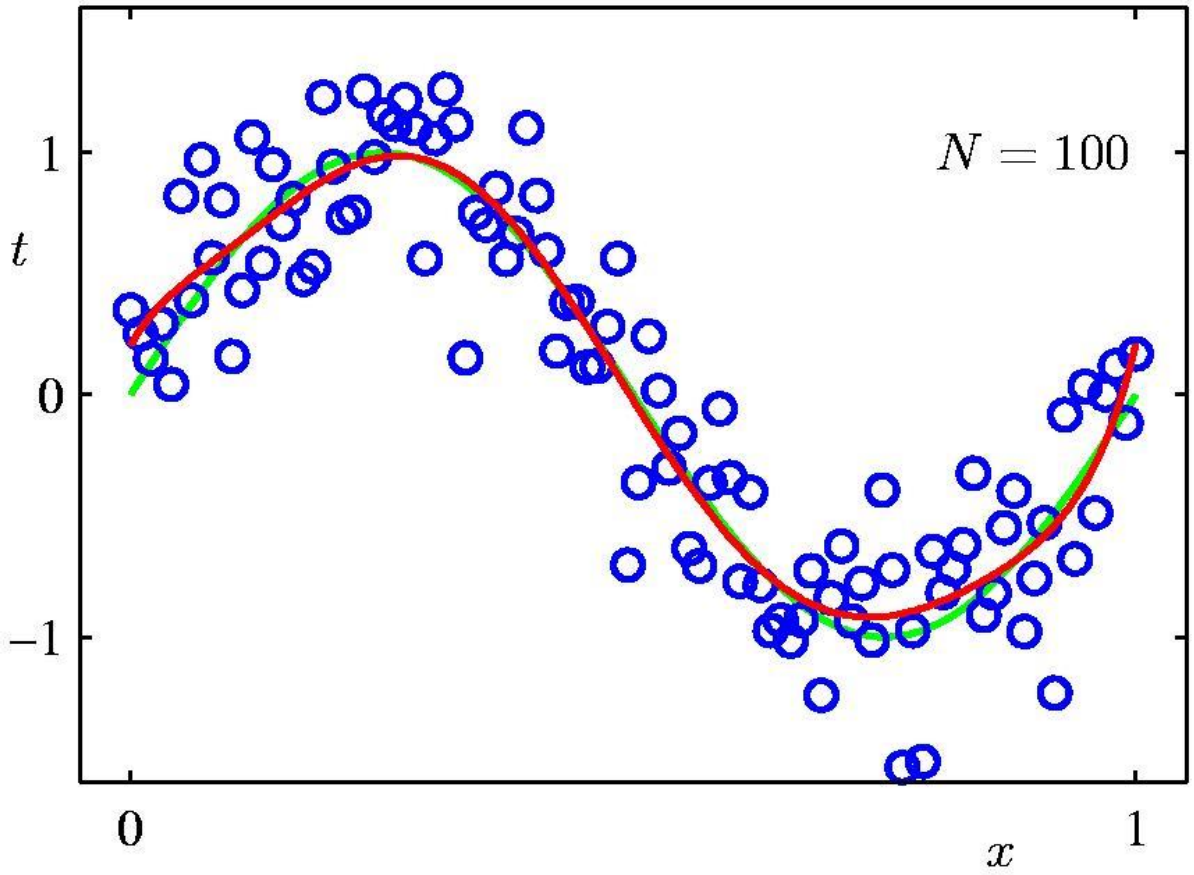
$N = 15$



9th Order Polynomial

When the training data set size N varies

$N = 100$



9th Order Polynomial

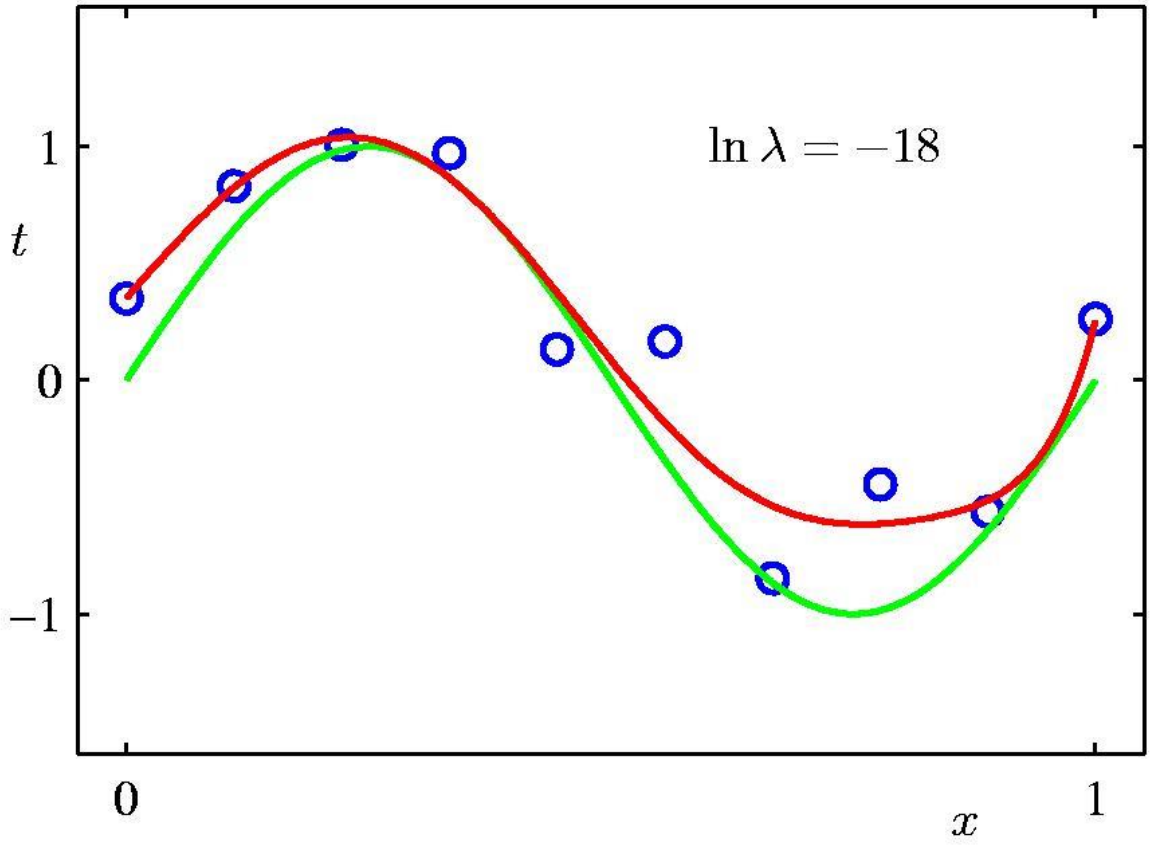
Considering Regularization

- Penalize large coefficient values

$$\widetilde{E}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$$

Under Different Regularization strengths

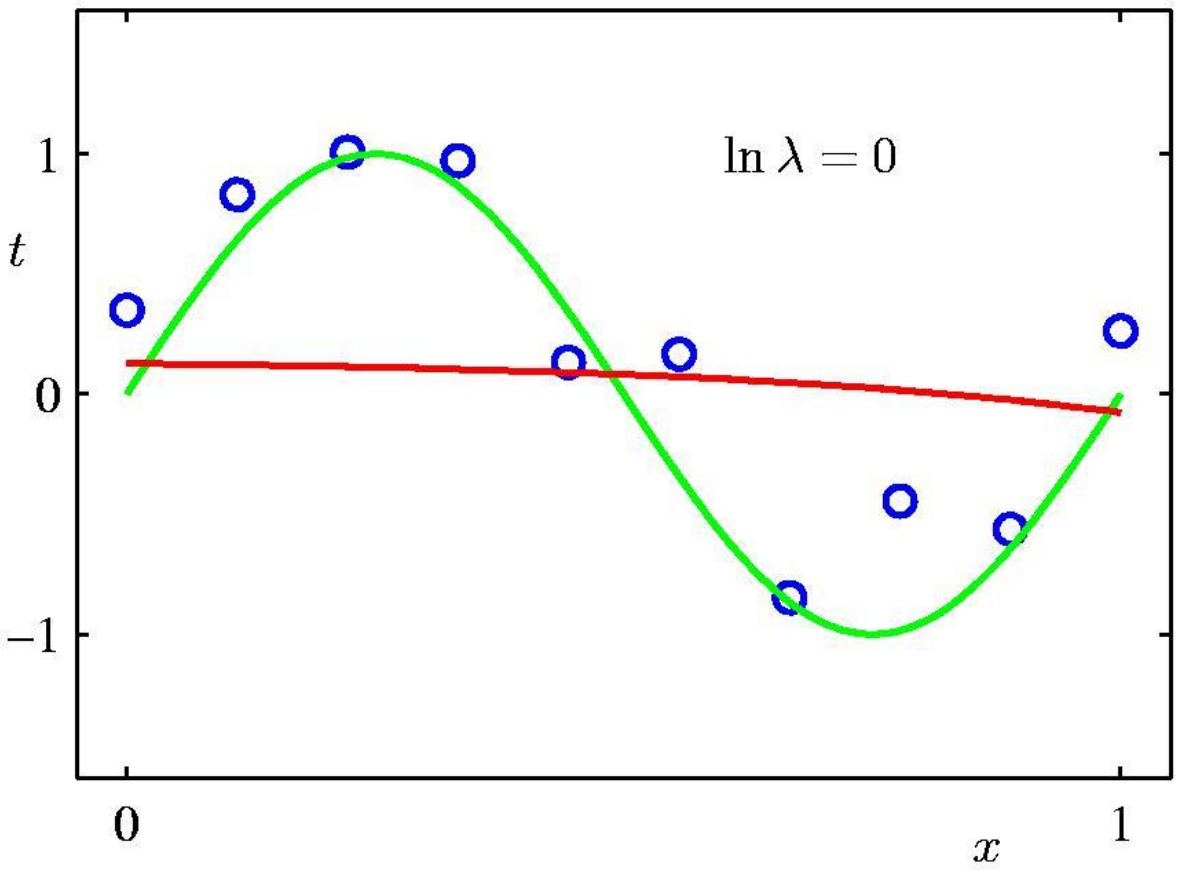
$$\lambda = e^{-18}$$



$$\widetilde{E}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$$

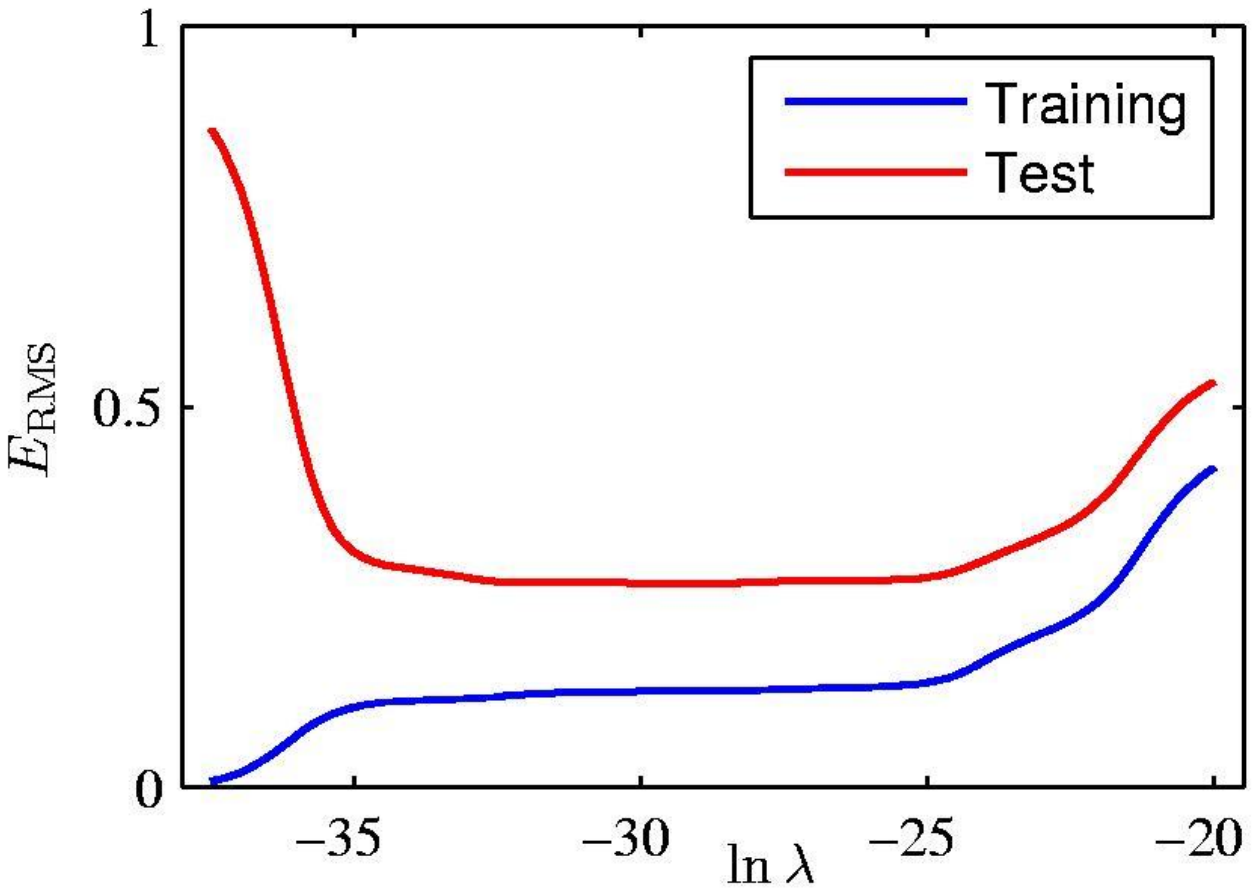
Under Different Regularization strengths

$$\lambda = e^0 = 1$$



$$\widetilde{E}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$$

Training and testing under different values of λ



$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

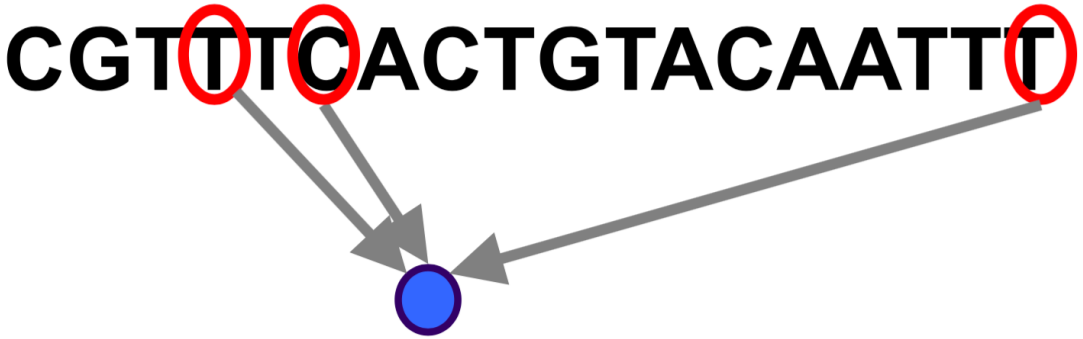
Polynomial Coefficients

	$\ln \lambda = -\infty$	$\ln \lambda = -18$	$\ln \lambda = 0$
$w_0^\star$	0.35	0.35	0.13
$w_1^\star$	232.37	4.74	-0.05
$w_2^\star$	-5321.83	-0.77	-0.06
$w_3^\star$	48568.31	-31.97	-0.05
$w_4^\star$	-231639.30	-3.89	-0.03
$w_5^\star$	640042.26	55.28	-0.02
$w_6^\star$	-1061800.52	41.32	-0.01
$w_7^\star$	1042400.18	-45.95	-0.00
$w_8^\star$	-557682.99	-91.53	0.00
$w_9^\star$	125201.43	72.68	0.01

Case study: predicting gene expression

The genetic picture

causal SNPs



a univariate phenotype:

i.e., the expression intensity of a gene

Association Mapping as Regression

	Phenotype (BMI)	Genotype
Individual 1	2.5	..C.....T..C.....T... ..C.....A..C.....T...
Individual 2	4.8	..G.....A..G.....A... ..C.....T..C.....T...
⋮		
Individual N	4.7	..G.....T..C.....T... ..G.....T..G.....T...

Benign SNPs

Causal SNP

Association Mapping as Regression

	Phenotype (BMI)	Genotype
Individual 1	2.5	. . 0 . . . . . 1 . . 0 . . . . . 0 . . .
Individual 2	4.8	. . 1 . . . . . 1 . . 1 . . . . . 1 . . .
⋮		
Individual N	4.7	. . 2 . . . . . 2 . . 1 . . . . . 0 . . .

$$y_i = \sum_{j=1}^{J} x_{ij}\beta_j$$

SNPs with large $|\beta_j|$ are relevant

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